

```

!pip install gradio
import gradio as gr
import random

def predict_electricity_bill(usage_kwh, state, consumer_type):
    # Base rate per kwh (fictional data)
    base_rates = {
        "Karnataka": 5.5,
        "Maharashtra": 6.0,
        "Tamil Nadu": 5.8,
        "Delhi": 6.2,
        "Gujarat": 5.7
    }

    base_rate = base_rates.get(state, 6.0) # Default to 6.0 if state not found

    # Adjust rate based on consumer type
    if consumer_type == "Residential":
        base_rate *= 1.0
    elif consumer_type == "Commercial":
        base_rate *= 1.2
    else: # Industrial
        base_rate *= 1.5

    # Calculate base amount
    base_amount = usage_kwh * base_rate

    # Calculate additional charges
    fixed_charge = random.uniform(50, 150)
    fuel_surcharge = base_amount * 0.05
    regulatory_charge = base_amount * 0.02

    # Calculate GST (assuming 18% for electricity)
    subtotal = base_amount + fixed_charge + fuel_surcharge + regulatory_charge
    gst = subtotal * 0.18

    # Calculate total bill
    total_bill = subtotal + gst

    # Generate energy-saving tip
    tips = [
        "Use LED bulbs to save up to 80% energy compared to traditional bulbs.",
        "Turn off appliances when not in use to avoid phantom power consumption.",
        "Use natural light whenever possible to reduce lighting costs.",
        "Set your air conditioner to 24°C for optimal energy efficiency.",
        "Use solar panels to reduce your dependence on grid electricity."
    ]
    energy_tip = random.choice(tips)

    # Format the output
    output = f"""
    Electricity Bill Prediction:

    Base Amount: ₹{base_amount:.2f}
    Fixed Charge: ₹{fixed_charge:.2f}
    Fuel Surcharge: ₹{fuel_surcharge:.2f}
    Regulatory Charge: ₹{regulatory_charge:.2f}

    Subtotal: ₹{subtotal:.2f}
    GST (18%): ₹{gst:.2f}

    Total Bill: ₹{total_bill:.2f}

    Energy Saving Tip: {energy_tip}

    Note: This is a predicted bill based on average rates and may not reflect your actual bill.
    Please check with your local electricity board for accurate billing information.
    """

    return output

# Create the Gradio interface
# Removed the layout parameter as it's not supported in gr.Interface
iface = gr.Interface(
    fn=predict_electricity_bill,
    inputs=[
        gr.Slider(minimum=0, maximum=1000, step=10, label="Electricity Usage (kWh)"),
        gr.Dropdown(["Karnataka", "Maharashtra", "Tamil Nadu", "Delhi", "Gujarat"], label="State"),
        gr.Radio(["Residential", "Commercial", "Industrial"], label="Consumer Type")
    ],
    outputs=gr.Textbox(label="Predicted Electricity Bill"),

```

```
title="India Electricity Bill Predictor",
description="Estimate your electricity bill based on usage, state, and consumer type.",
theme="huggingface",
examples=[
    [250, "Karnataka", "Residential"],
    [500, "Maharashtra", "Commercial"],
    [750, "Delhi", "Industrial"]
]
)

iface.launch()
```

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